

BACKTEST PARAMETER COMPARISON

Screening yield, confirmation coverage, and parameter sensitivity

Evaluation window: 2022-01-26 to 2026-06-05 | Parameter sets: 192 | Compact outcomes: 5,626

Key conclusion: **volume-ratio threshold** is the dominant driver of screening yield. The broadest configuration selected 92 A-F and 47 A-H securities, while stricter volume settings reduced the candidate pool rapidly. This report compares screening coverage, not realized investment-return quality.

Parameter Choice Reference

- Six selectable parameters form the 192-combination grid. Daily and weekly moving-average tolerances remain fixed across every parameter set.

Parameter	Conditions	Choices	Mode	Meaning
Annual growth %	A	2%, 3%	Selectable	Minimum annual revenue and operating-income growth.
Quarterly growth %	B	2%, 3%	Selectable	Minimum quarterly revenue and operating-income growth.
Annual periods	A	2, 3 years	Selectable	Number of latest annual periods that must pass.
Quarterly periods	B	2, 3, 4 quarters	Selectable	Number of latest quarterly periods that must pass.
Volume ratio	C / D	2x, 3x, 4x, 5x	Selectable	Required volume-surge multiple.
Volume surge days	D	2, 3 days	Selectable	Minimum surge days in the recent-volume window.
Daily MA tolerance	E / F	1%	Fixed	Allowed daily MA20/MA50/MA100 alignment tolerance.
Weekly MA tolerance	G / H	2%	Fixed	Allowed weekly MA5/MA10/MA30 alignment tolerance.

Executive Summary

- The run completed successfully across all 192 parameter combinations with no duplicate outcomes, missing core prices, invalid dates, or high/low inconsistencies.
- A-F produced 3,698 parameter-security outcomes across 192 parameter sets and 92 unique securities.

- A-H produced 1,928 outcomes across 192 parameter sets and 47 unique securities; weekly confirmation narrows the candidate universe.
- Lower volume-ratio thresholds and shorter fundamental-history requirements produce the largest candidate pools.
- Return, median return, win rate, and drawdown ranking are not present in the execution log; those metrics should be queried from `backtest\_selection\_outcome` before selecting a final strategy.

Validated Run Overview

Screen	Outcomes	Sets with results	Unique stocks	First signal	Last entry
A-F	3,698	192	92	2022-01-26	2026-05-28
A-H	1,928	192	47	2022-01-26	2026-06-05

Average selected stocks per parameter set

A-F and A-H screening yield by selected parameter value



Figure 1. Average screening yield by parameter choice.

Parameter Sensitivity

Factor	Compared values	Avg A-F yield	Avg A-H yield
Annual growth threshold	2 -> 3	19.9 -> 18.6	10.5 -> 9.6
Quarterly growth threshold	2 -> 3	20.1 -> 18.5	10.4 -> 9.7

Annual periods	2 -> 3	26.2 -> 12.4	13.7 -> 6.4
Quarterly periods	2 -> 4	23.6 -> 15.3	12.3 -> 8.1
Volume ratio threshold	2 -> 5	51.4 -> 3.6	25.5 -> 3.1
Volume surge days	2 -> 3	22.1 -> 16.4	11.4 -> 8.7

Interpretation

- Volume ratio: moving from 2 to 5 changes average A-F yield from 51.4 to 3.6 and average A-H yield from 25.5 to 3.1.
- Annual-period requirement: moving from 2 to 3 changes average A-F yield from 26.2 to 12.4 and average A-H yield from 13.7 to 6.4.
- Quarterly-period requirement: moving from 2 to 4 changes average A-F yield from 23.6 to 15.3 and average A-H yield from 12.3 to 8.1.
- Volume-surge-day requirement: moving from 2 to 3 changes average A-F yield from 22.1 to 16.4 and average A-H yield from 11.4 to 8.7.

Highest-Coverage Parameter Sets

ID	Growth A/Q	Periods A/Q	Volume	A-F	A-H	Total
1	2/2	2/2	2x/2d	92	47	139
97	3/2	2/2	2x/2d	88	44	132
49	2/3	2/2	2x/2d	85	45	130
145	3/3	2/2	2x/2d	81	41	122
9	2/2	2/3	2x/2d	79	41	120
2	2/2	2/2	2x/3d	78	37	115
105	3/2	2/3	2x/2d	76	38	114
57	2/3	2/3	2x/2d	73	38	111
50	2/3	2/2	2x/3d	73	37	110
98	3/2	2/2	2x/3d	74	34	108

Lowest-Coverage Parameter Sets

- These configurations produced the fewest combined A-F and A-H selections. Their sparse samples are useful for understanding selectivity, but are too small for reliable return-performance conclusions.

ID	Growth A/Q	Periods A/Q	Volume	A-F	A-H	Total
46	2/2	3/4	4x/3d	1	1	2
47	2/2	3/4	5x/2d	1	1	2
48	2/2	3/4	5x/3d	1	1	2
87	2/3	3/3	5x/2d	1	1	2
88	2/3	3/3	5x/3d	1	1	2
94	2/3	3/4	4x/3d	1	1	2
95	2/3	3/4	5x/2d	1	1	2
96	2/3	3/4	5x/3d	1	1	2
135	3/2	3/3	5x/2d	1	1	2
136	3/2	3/3	5x/3d	1	1	2

Recommended Comparison Set

Role	ID	Configuration	A-F / A-H	Use
Broad discovery	1	2%/2%, 2 annual, 2 quarters, 2x volume, 2 surge days	92 / 47	Largest sample; suitable baseline.
Moderate volume	3	2%/2%, 2 annual, 2 quarters, 3x volume, 2 surge days	36 / 18	Stricter volume comparator.
Strict volume	7	2%/2%, 2 annual, 2 quarters, 5x volume, 2 surge days	9 / 7	High-conviction, smaller-sample comparator.
Durability focus	41	2%/2%, 3 annual, 4 quarters, 2x volume, 2 surge days	33 / 18	Longer fundamental-history comparator.

Decision Guidance

- Do not label the highest-coverage parameter set as the best-performing investment strategy. Coverage and return quality answer different questions.

- Require a minimum sample size before ranking return performance. A practical initial threshold is at least 20 A-F outcomes or at least 10 A-H outcomes.
- Compare mean and median return together. A large gap often indicates that a few extreme winners dominate the average.
- Use win rate and maximum drawdown alongside returns; parameter sets with high average returns but severe drawdowns may be unsuitable.
- Evaluate A-F and A-H separately. A-H is a stricter weekly-confirmed strategy and should not be pooled with A-F.

Required Return-Performance Query

To complete an investment-performance ranking, export the following aggregate query from Supabase and use it for a second report:

```
SELECT p.parameter_set_id, p.parameter_set_name, o.screen_type, COUNT(*) AS sample_size,
ROUND(AVG(o.return_pct), 2) AS avg_return_pct,
ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY o.return_pct)::numeric, 2) AS median_return_pct,
ROUND(100.0 * COUNT(*) FILTER (WHERE o.return_pct > 0) / COUNT(*), 2) AS win_rate_pct,
ROUND(AVG(o.max_drawdown_pct), 2) AS avg_drawdown_pct,
ROUND(AVG(o.max_return_pct), 2) AS avg_max_return_pct
FROM backtest_parameter_set p JOIN backtest_selection_outcome o USING (parameter_set_id)
GROUP BY p.parameter_set_id, p.parameter_set_name, o.screen_type
ORDER BY o.screen_type, sample_size DESC;
```

Scope and Limitations

- This report uses the validated execution log and compares selection yield, not realized portfolio returns.
- The same security may appear in many parameter sets, so outcome rows are not independent observations.
- Fundamental histories use `datadate <= signal\_date`; actual public filing-availability dates are unavailable, so some look-ahead-bias risk remains.
- No transaction costs, liquidity constraints, holding rules, or overlapping-position controls are included.